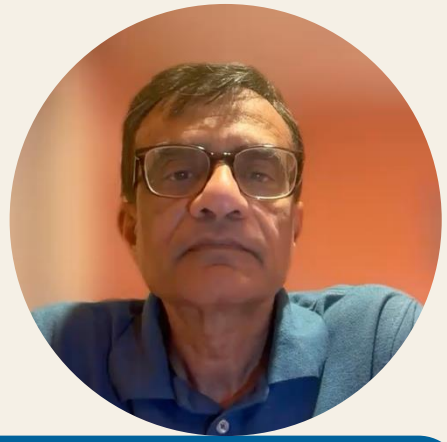
Embedded System HW Design
Week6
Internet Of Things Systems

- **IoT System Applications**
- **IoT System Architectures**
- **OSI model for networks**
- **Internet Protocol**
- **IoT Networking Concepts**
- **Example networks**:
  - Classic Bluetooth, Bluetooth Low Energy.
  - Wi-Fi.



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IoT System Applications

- **Soft real-time networked embedded system**.
  - Input devices: tags, sensors, etc.
  - Output devices: motor controllers, displays, etc.

- **Basic Flow** :
  Sense → Process → Communicate → Act
- **Examples**:
  - Computer-readable identification code for objects.
  - Appliances controlled by cell phone interface.
  - Sensor network with analytics.

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IoT Devices

- **People**:
  - Implanted devices in the body.
  - Wearable devices on the body.
  - Environmental devices outside the body.
- **Objects**:
  - Interior: temperature sensor, etc.
  - Exterior: RFID, etc.
  - Environmental: camera, motion sensor, etc.

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IoT RFID

- RFID tag can provide object ID (Electronic Product Code, etc.), other information.
- Many tags are read-only, some are writable.
- **Two types of tags**:
  - **Passive** transmits only when it receives a request.
  - **Active** tag both transmits independently and responds to requests.
- **Passive** may also be used to refer to tags with no internal power source.
- RFID tags may operate in several different bands and at different ranges.

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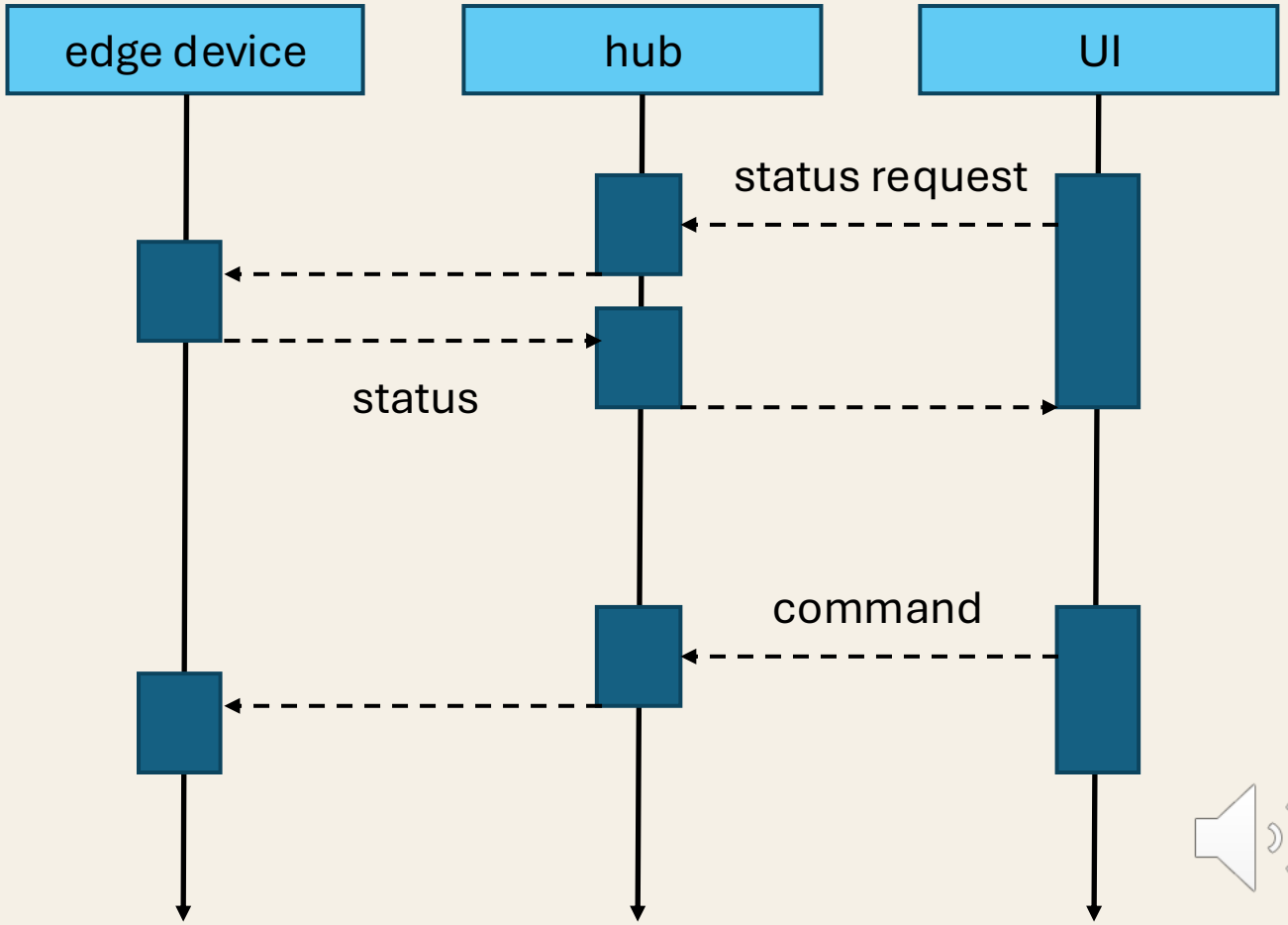
IoT System Architecture

- **Edge**: I/O devices.
- **Cloud**: centralized processing.
- **Smart appliance** = connected appliance + network + UI.



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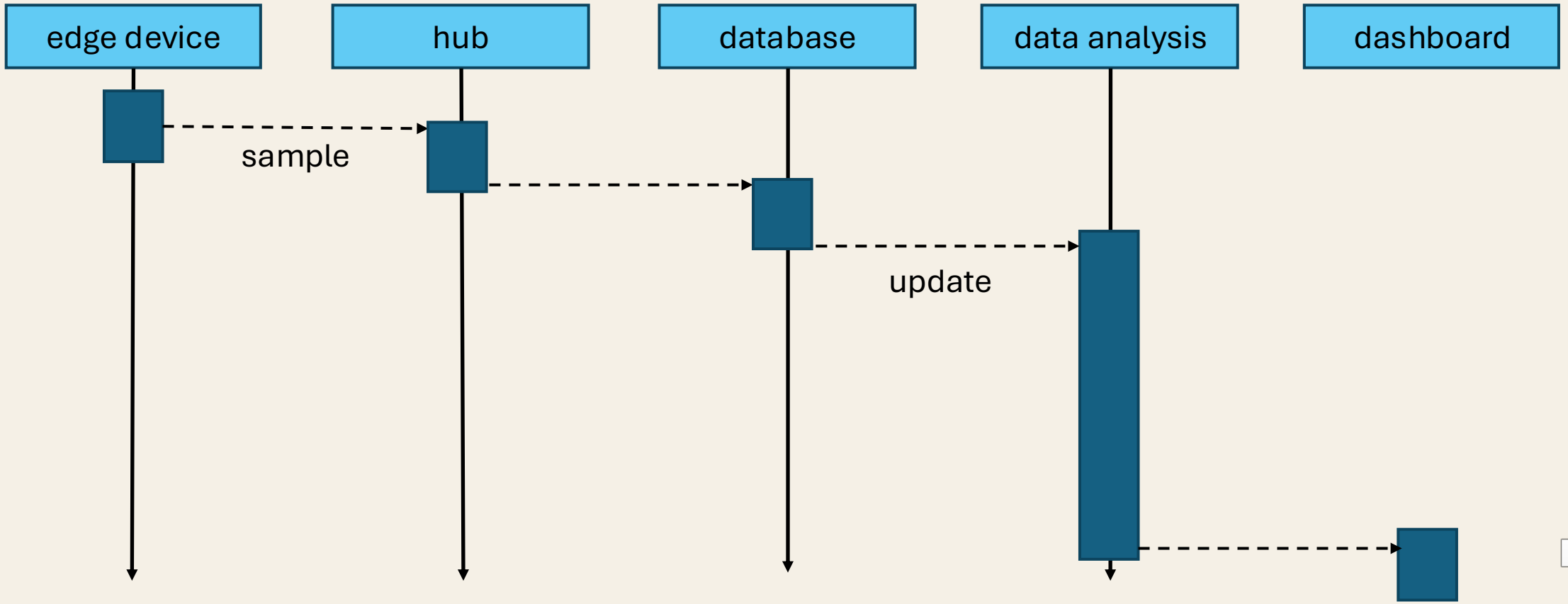
IoT System Architecture (Cont'd)

- **Monitoring system** = sensors + network + database + dashboard.



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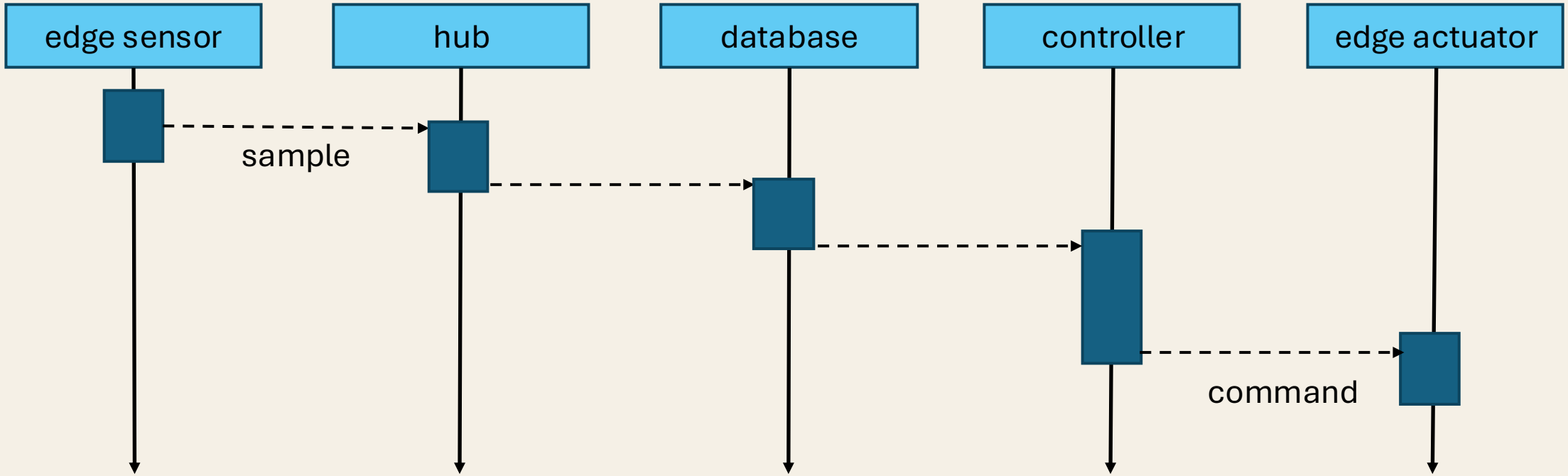
IoT System Architecture (Cont'd)

- **Control system** = sensors + database + controller + actuator.



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IoT System OSI Model

application	end-use interface
presentation	data format
session	application dialog control
transport	connections
network	end-to-end service
data link	reliable data transport
physical	mechanical, electrical

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IoT System OSI Layers

- **Physical**: connectors, bit formats, etc.
- **Data link**: error detection and control across a single link (single hop).
- **Network**: end-to-end multi-hop data communication.
- **Transport**: provides connections; may optimize network resources.
- **Session**: services for end-user applications: data grouping, checkpointing, etc.
- **Presentation**: data formats, transformation services.
- **Application**: interface between network and end-user programs

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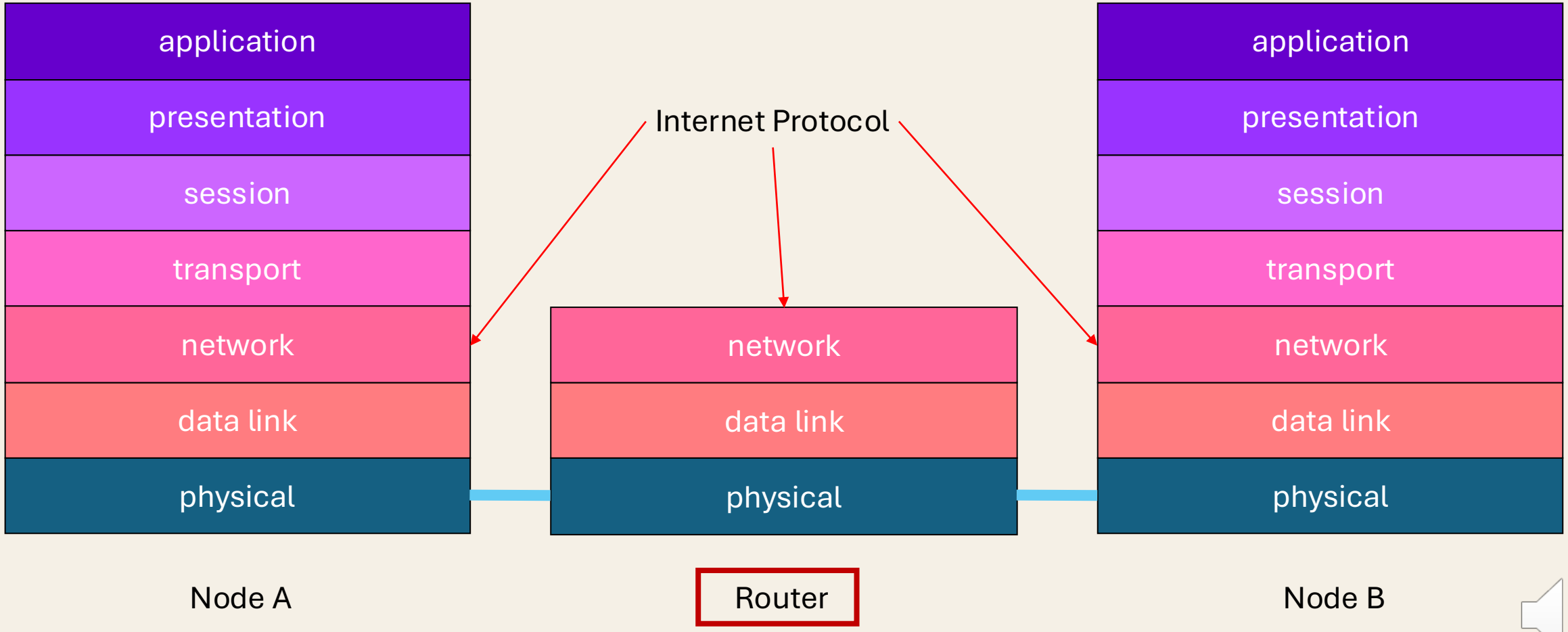
Internet Protocol



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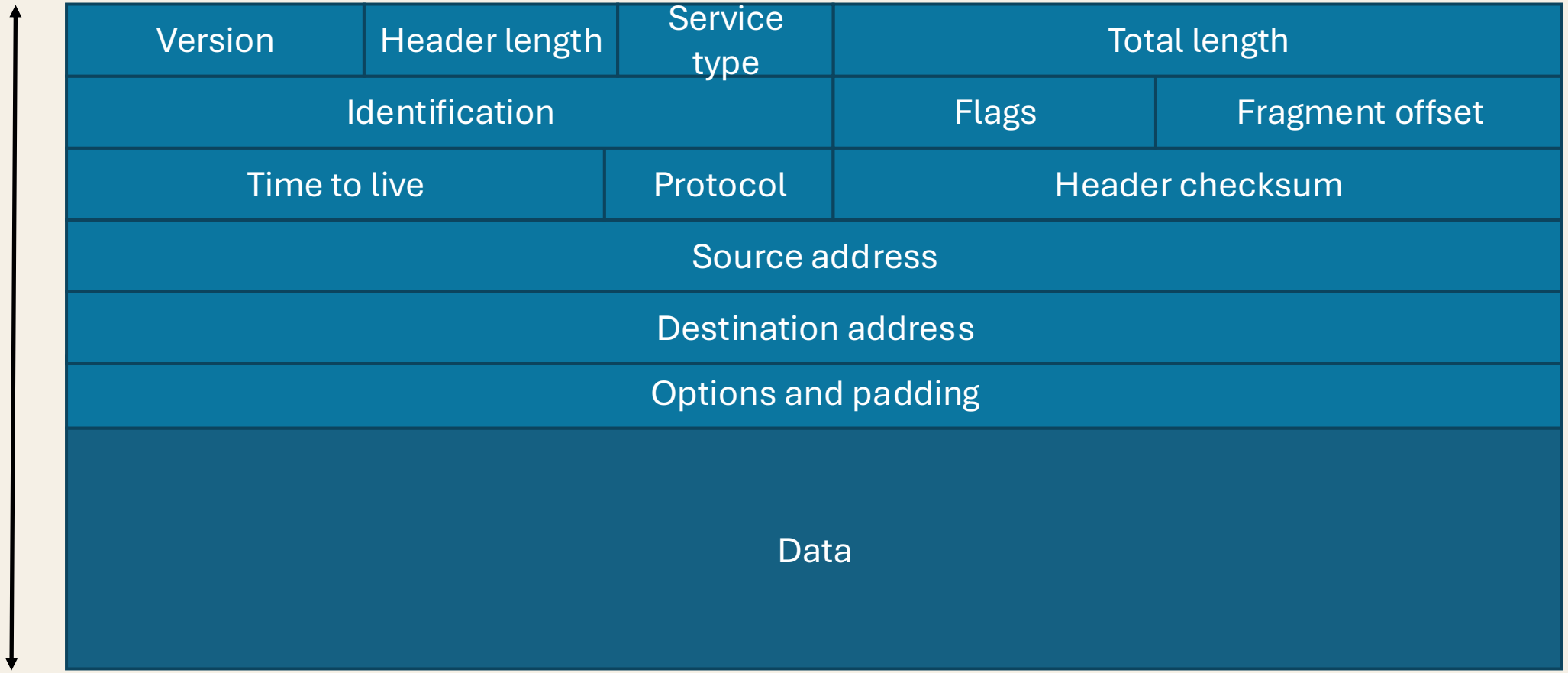
Internet Protocol

max 65,535 bytes



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IP Routing

- IP routing is best effort---no guarantees of packet delivery.
- Build other services on top of IP:
  - Use handshakes to verify delivery of packet.
  - Network routers can enforce quality-of-service.

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Internet Services

- Domain name service (DNS):
  - Map names onto IP addresses.
- File transfer (FTP):
  - Move files from machine to machine.
- Terminal sessions:
  - Telnet provides terminal-style access.
- Web (HTTP):
  - Built on top of FTP.
- Email (SMTP):
  - Built on top of FTP.

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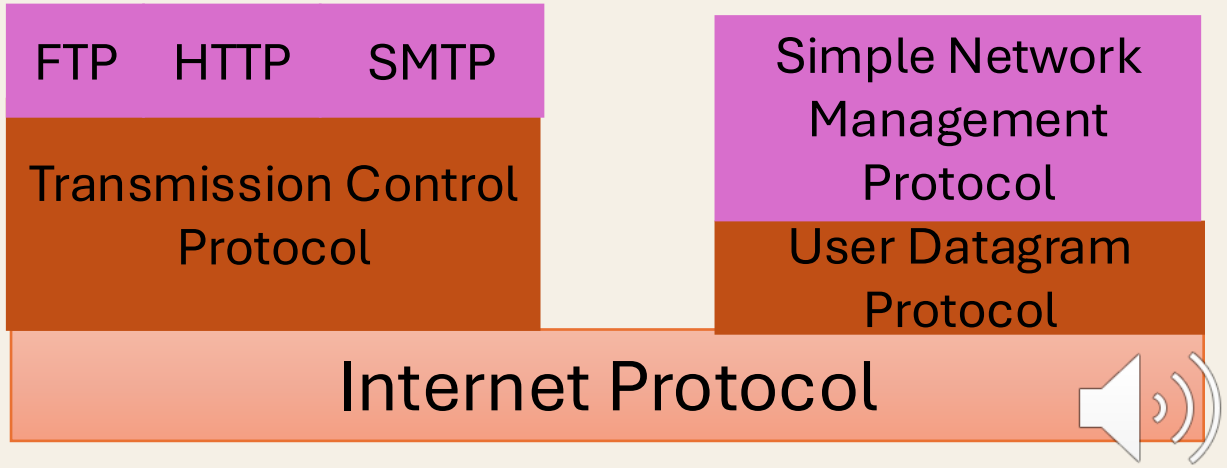
Host to Host Service

- **Hides some details of IP**:
  - Break host communication into IP packets at source.
  - Reassemble packets at destination.
  - Use handshake to ensure packets arrive, retransmit if necessary.

- **Transmission Control Protocol (TCP)**:
  - Connection-oriented service.

- **User Datagram Protocol (UDP)**:
  - Datagram service.
  - Datagram is modeled after a telegram.



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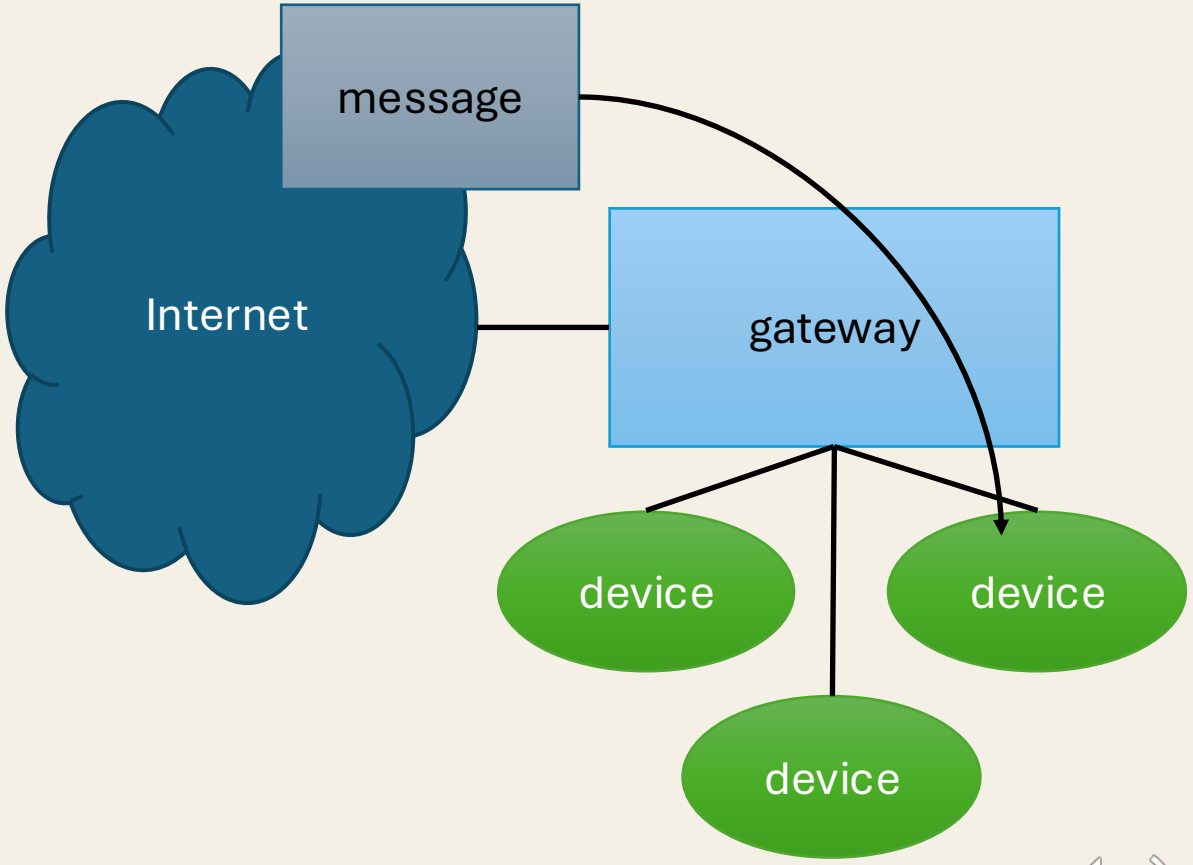
IoT Networking Concepts

- Edge device may not run IP protocol
  - IP connection may be provided by hub or gateway.
  - Non-IP networks are known as edge networks.

- Ad hoc network is self-organized---not set up by system administrator

- **Ad hoc network services**:
  - Authentication of eligibility to join network.
  - Authorization for access to given pieces of information on the network.
  - Encryption and decryption.



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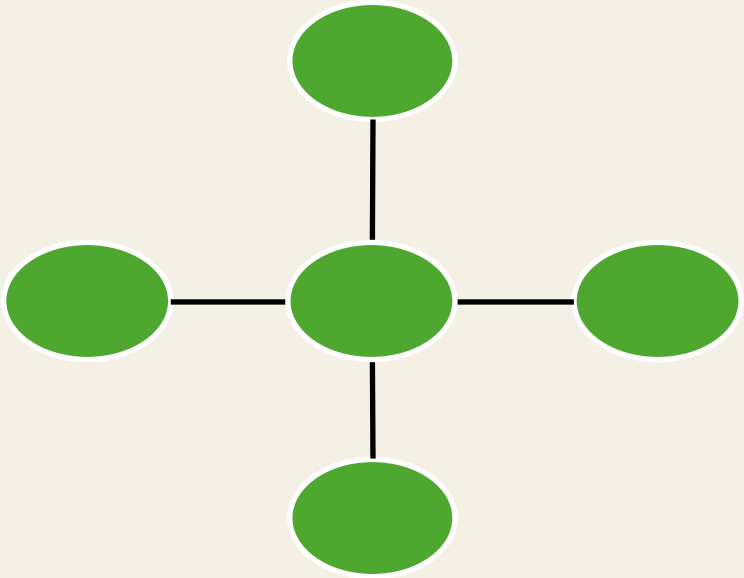
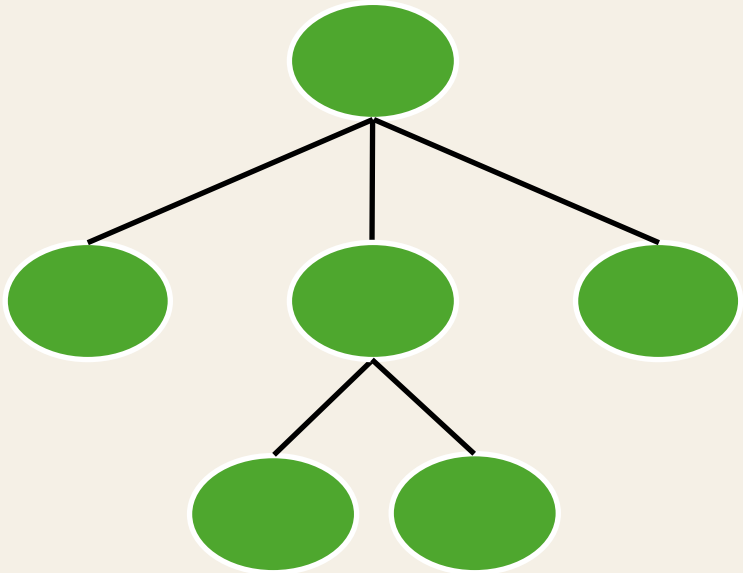
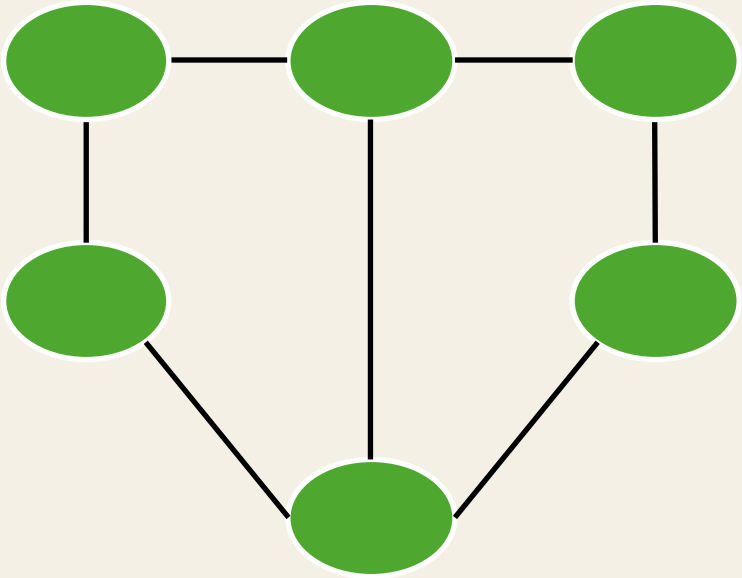
IoT Networking Topologies



star



tree



mesh

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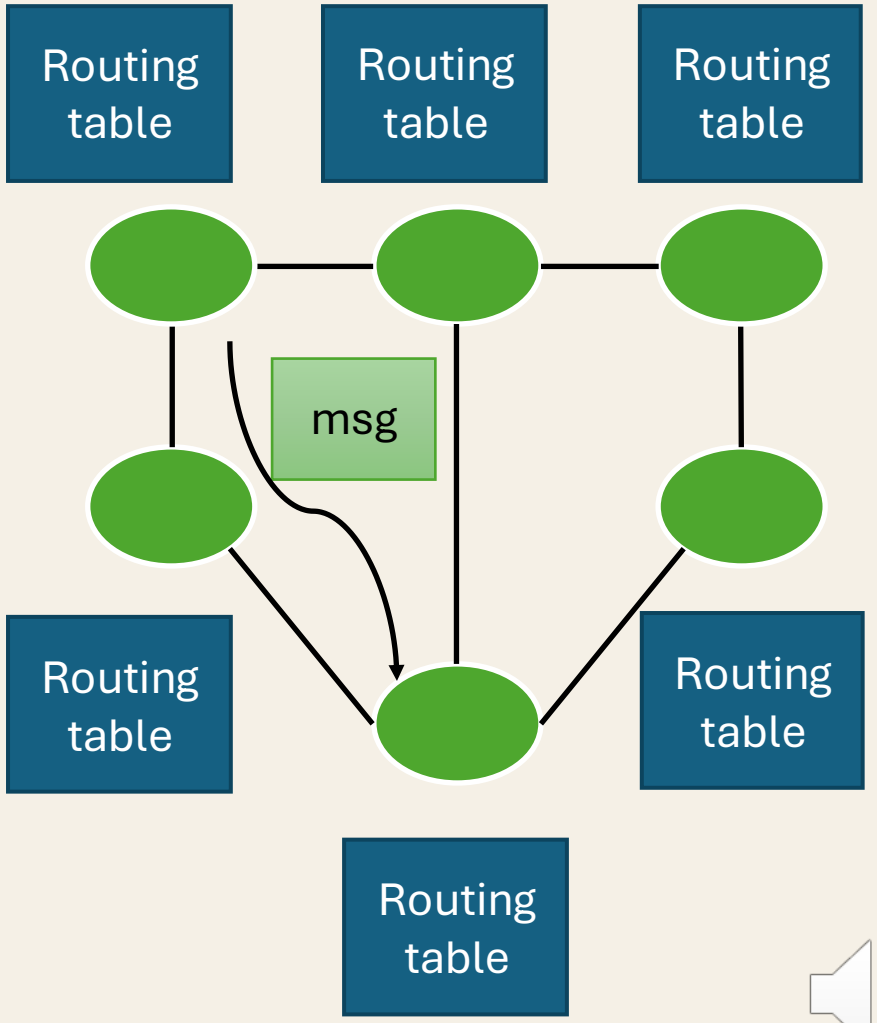
IoT Routing

- Routing discovery determines routes between source/destination pairs

- Routing is driven by routing tables at the nodes.



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IoT QoS

- Many networks support synchronous and asynchronous communication
  - Asynchronous: data records, etc.
  - Synchronous: voice, etc.

- **Quality-of-service (QoS):** bandwidth and periodicity characteristics

- Admission control ensures that network can handle the QoS demands of a request.

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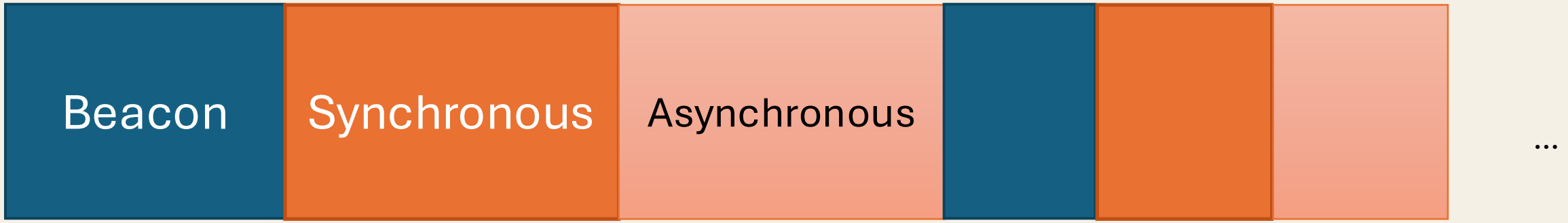
IoT Synchronization and beacons

- Many network operations require nodes to be synchronized

- Synchronization can be performed using Beacon.
  - Beacon transmission marks the beginning of a communications interval.



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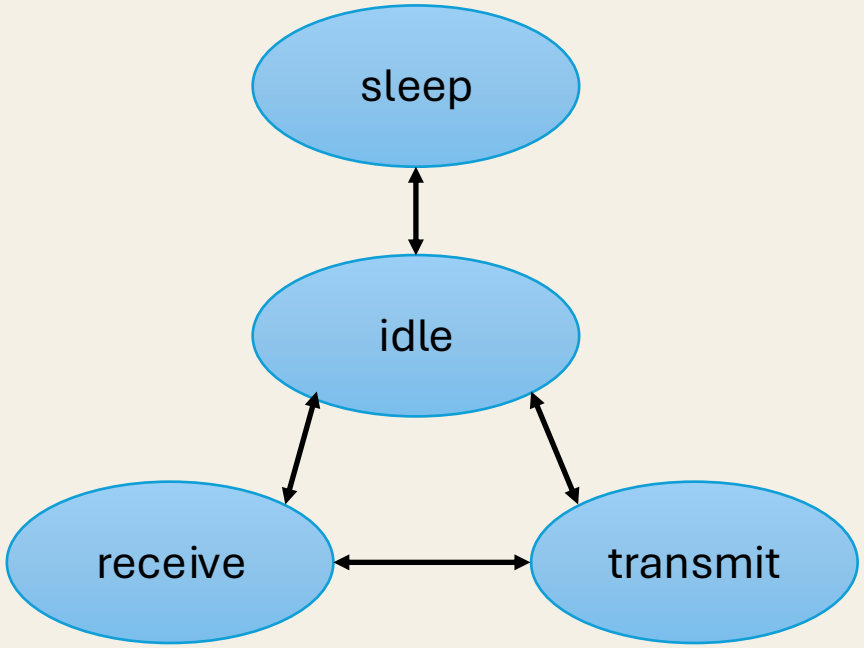
IoT Communications energy

- Communications energy is a large part of node energy consumption
- Comm energy consumption depends on many factors and parameters
  - Generally evaluated for a set of use cases.
- We can use power state machine to model communications energy cost



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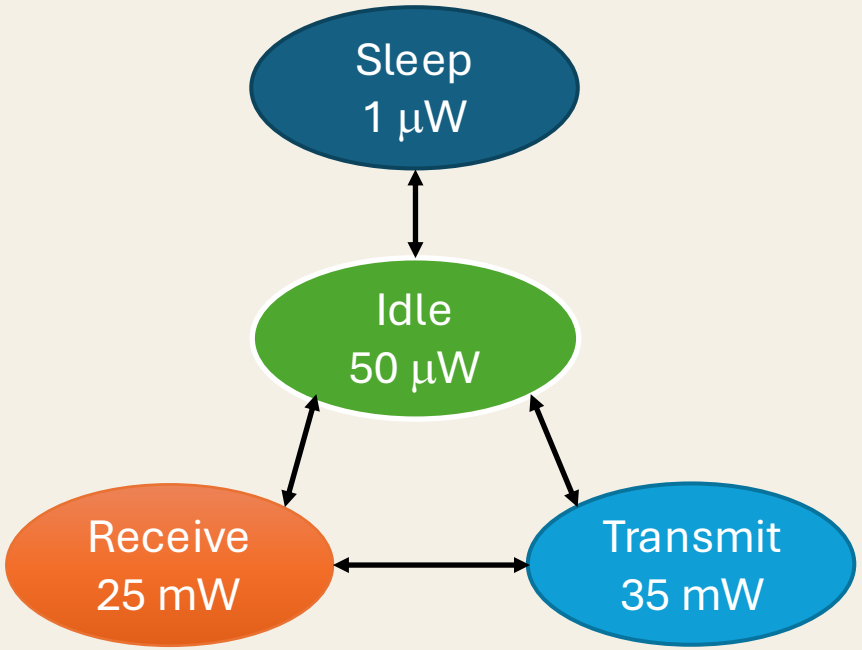
IoT Communications Power State Machine Example



step	state	time	energy
1	Sleep	1 ms	1 nJ
2	Idle	10 μs	0.5 nJ
3	Receive	50 μs	1.25 μJ
4	Transmit	50 μs	1.75 μJ
5	Receive	50 μs	1.25 μJ
6	Transmit	50 μs	1.75 μJ
			total = 6 μJ

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IoT Bluetooth

- Introduced in 1999, originally for telephony applications
- Classic Bluetooth operates in instrumentation, scientific, and medical (ISM) band in the 2.4 GHz range
- Bluetooth networks organized as piconet
  - One master, several slaves.
  - Slave can be active or parked.
  - A device can be a slave on several networks simultaneously.

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IoT Bluetooth Stack

- **Transport protocol**:
  - Radio, baseband layer, link manager, logical link control and adaptation protocol (L2CAP).

- **Middleware**:
  - RFCOMM for serial port, service discovery protocol, Internet Protocol, IrDA, etc.

- **Applications.**

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IoT Bluetooth Protocol

- Every Bluetooth device has a 48-bit Bluetooth Device Address
- Every device has a Bluetooth clock
- Transmissions alternate between master and slave directions
- **Two types of packets**:
  - Synchronous connection-oriented (SCO) packets for QoS-oriented traffic.
  - Asynchronous connectionless (ACL) packets for non-QoS traffic.
  - SCO traffic has higher priority than ACL packets.

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IoT Bluetooth Low Engergy(BLE)

- Designed for very low energy operation such as button-sized battery
  - Goal: minimize radio on-time.
- Part of Bluetooth standard but deviates from Classic Bluetooth in several ways
- Advertising transmissions can be used to broadcast, discover devices, etc.
- Connections can be established.
- Attribute Protocol Layer allows devices to create application-specific protocols.
- Generic Attribute Profile Layer (GATT) defines basic attributes for all BLUE devices.
- Pairing devices uses a short-term key to send a long-term key.
  - Bonding: storing long-term key in device database.
  - Optional data encryption using AES.

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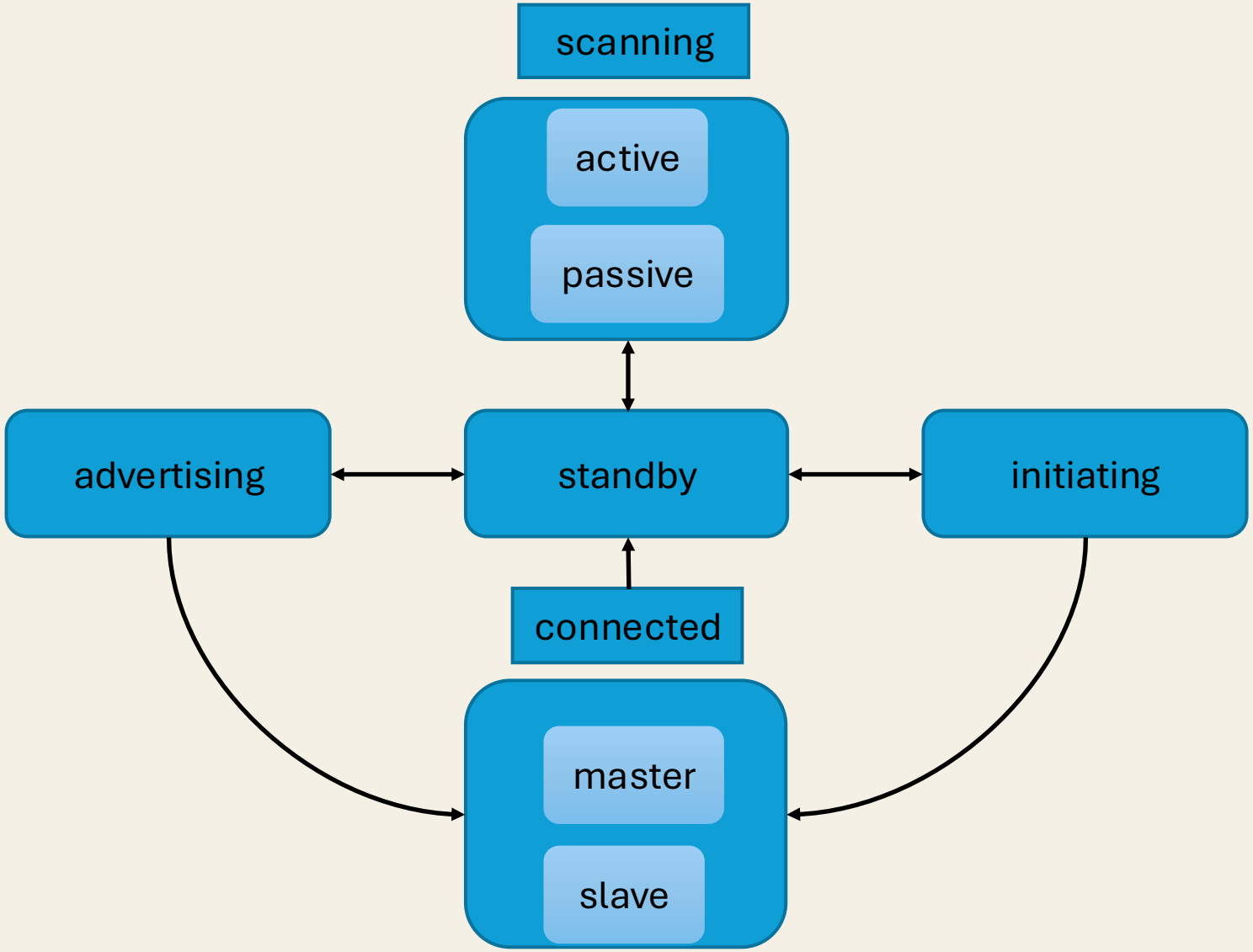
IoT BLE State Machine



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IoT Wi-FI

- Originally designed for portable and mobile applications
  - Has been adapted for lower-energy operation.
- Supports ad hoc networking
- **Network provides a set of services**:
  - Distribution of messages from one node to another.
  - Integration delivers messages from another network.
  - Association relates a station to an access point.

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